

MusicX: Fast distillation of Consistency Trajectory Models for X-to-Music Guided Generation

Rajalaxmi Rajagopalan^{1,**}, Martin Sipka², Erfan Soltanmohammadi², Turab Iqbal²

¹ University of Illinois, Urbana-Champaign

² AWS AI Labs

Abstract

Diffusion-based text-to-music (TTM) generators, although powerful tools for on-the-fly music composition and content creation, are mostly text-conditioned, or require expensive re-training or finetuning (applies to other occurrences too) to allow other conditioning modalities, are only capable of generating music of a certain fixed length, and have long inference times as they require a large number of diffusion steps or Function Evaluations (NFEs) to generate good-quality music. While existing methods like Consistency Models (CMs) and Consistency Trajectory Models (CTMs) reduce NFEs by distillation, either the quality is heavily degraded, or the distillation is prohibitively costly. We propose MusicX, a fast and versatile music generator that addresses these issues by, (1) *Guidance Control*: modifies the diffusion trajectories through an optimization in the diffusion latent space, enabling training-free modification for arbitrary conditioning modalities. (2) *Fused Diffusion*: merges multiple guided diffusion trajectories, enabling variable-length multi-conditioned generation. (3) *Fast Distill*: speeds up the distillation by several orders of magnitude by modifying the CTM distillation scheme. Our evaluations show that our method achieves SOTA performance in terms of generation speed (NFEs), music quality (FAD), and adherence to input controls (CLAP, LPAPS, MSE), over comparable approaches in the literature.

Index Terms: Music Generation, Inference-Time Optimization, Diffusion Model Distillation

1. Introduction

Diffusion model continues to make remarkable advances in high-quality and scalable generation of a wide range of content, such as images [1], video [2, 3], audio [4], protein structures [6], etc. Latent Diffusion Models (LDMs) like AudioLDM [7], AudioLDM2 [4], and Stable Audio [8] perform diffusion in a compressed latent space and offer user-controllability through class-conditional generation. Text-to-Music (TTM) diffusion models like MusicLDM [5] offer state-of-the-art performance in terms of music generation quality and text-based global stylistic guidance. However, adapting user controllability to new modalities, variable length generation, and generation speed vs quality trade-off remain open challenges, mostly addressed by expensive re-training and fine-tuning.

More recently, there is growing interest in developing training-free inference-time optimization schemes for more fine-grained control. Inference-time optimization works [9–12] modify the latent input and/or diffusion parameters of popular LDMs like AudioLDM2 [4] to steer the generation process along preferred directions. DITTO [11] optimizes the initial noise latent

(seed) through a differentiable feature matching loss. Authors demonstrate competitive generation performance for a wide variety of tasks, including inpainting, outpainting, and looping, as well as music features like intensity, melody, and musical structure control, but with an increase in inference time and Number of Function Evaluations NFEs. This is remedied by model distillation techniques like Consistency Models (CMs) [13] and Consistency Trajectory Models (CTMs) [14], which offer a significant speed-up by drastically reducing the NFEs from about 100 (DDIM [15]) to 1-5. DITTO-2 [12] employs a Consistency model (CM) that offers one-step generation (NFE=1) at the expense of quality. However, the flexibility of adjusting the generated quality by increasing the steps or NFEs is not possible in CMs. Hence, DITTO-2 underutilizes the inference-time optimization scheme by restricting it to only seed selection and doesn't explore more complex conditioning or controls that might require optimization over several steps. SoundCTM [16] uses a Consistency Trajectory Model (CTM), which offers flexible NFE choice. However, CTM distillation is computationally expensive, almost reaching the same order of magnitude required for training an LDM from scratch. SoundCTM also uses complex distillation training objectives like using a teacher LDM model's layers as feature extractors for computing the distillation distance, which impacts its generalizability. Moreover, in these works Variable-length generation is often restricted to the outpainting task given a reference audio [10, 11].

In this work, we develop a music generator that addresses the open challenges on generation controllability, length, quality, speed, and bridges the gap in the literature. MusicX generates music that is (1) conditioned on a wide range of input modalities like image, text, audio, symbolic music like MIDI, sheet music score, by performing optimization at multiple diffusion time points, (2) controlled by a variety of controls like intensity, rhythm, melody, music structure, specific genre or instrument, and emotion, and (3) of variable length with arbitrary segments following different control or conditioning by fusing multiple diffusion paths. Lastly, MusicX achieves near real-time generation speeds and tunable quality by training a CTM with a modified training scheme (following [17]). The new training scheme not only reduces training speed but also allows the use of Classifier-Free Guidance (CFG) and higher-order ODE-based samplers with the CTM.

2. Proposed Method

The forward process in a diffusion model generates progressively noisier versions of a clean sample x_0 based on a noise schedule $\{\alpha_i\}_{i=0}^T$ as follows,

$$x_t = \sqrt{\alpha_t} z_0 + \sqrt{1 - \alpha_t} \epsilon \quad \epsilon \in \mathcal{N}(0, \mathbb{I}) \quad (1)$$

^{**} indicates the corresponding author.

Latent Diffusion Models (LDMs) achieve computational efficiency by splitting the diffusion model training objective into a perceptual compression stage using a VAE, and a semantic compression stage, in which the actual diffusion model learns the semantic and conceptual composition of the data (distribution over music) [1]. The LDM latent space $\mathcal{Z}$ is thus a perceptually equivalent space to the input space $\mathcal{X}$. The LDM noise prediction model is typically a UNet [1] denoiser ϵ_θ that estimates the noise added at each timestep t ,

$$\epsilon_\theta \approx \epsilon = \frac{z_t - \sqrt{\alpha_t} z_0}{\sqrt{1 - \alpha_t}} \quad (2)$$

The data prediction model predicts the one-step clean sample $\hat{z}_0$ based on the noise estimate ϵ_θ .

$$\hat{z}_0 = \frac{z_t - \sqrt{1 - \alpha_t} \epsilon_\theta(z_t, t)}{\sqrt{\alpha_t}} \quad (3)$$

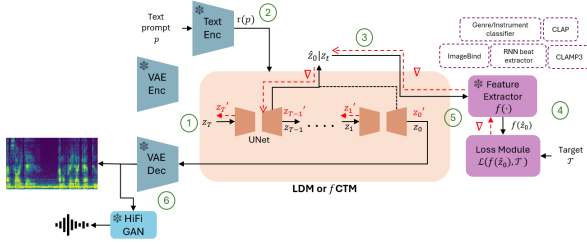


Figure 1: *MusicX*: (1) Initial noise latent z_T and (2) input text embedding $\tau(p)$ to the denoiser (TTM model). (3) The predicted clean sample, $\hat{z}_0|z_t$, is passed to the guidance function $f(\cdot)$. (4) Loss is computed w.r.t a target $\mathcal{T}$ and (5) optimizer computes the gradient update that modifies diffusion trajectory z_t to $z'_t \forall t$. (6) The edited clean sample z'_0 is converted to a raw waveform.

All samplers produce z_{t-1} based on current sample z_t , the current time step t , and the predicted noise $\hat{\epsilon}$ as $z_{t-1} = S(z_t, \hat{\epsilon}, t)$. User-defined controls - typically text - are applied to music generation by training Latent Diffusion Models (LDMs) on text-music pairs. The sampling function is modified to accept a control signal c as $z_{t-1} = S(z_t, \hat{\epsilon}, t, c)$. Thus, expensive re-training is necessary when the control type changes. In *MusicX*, we frame any modifications to the generation as an optimization in the latent space $\mathcal{Z}$ of the LDM. To generate a new sample $\tilde{z}_0$ that fits a particular user requirement is the same as identifying the update ∇ such that $\tilde{z}_0 = z_0 + \nabla$ i.e., by introducing an auxiliary optimization in the LDM latent space, we can modify the functionality of any base LDM. The optimization in *MusicX* has a dual objective: enable multi-modality conditioning and variable length generation. It is obvious that latent optimization incurs the cost of additional Denoising Network Function Evaluations (NFEs), which in turn increases generation time. It is imperative to explore NFE-reducing techniques that modify the sampling method, the LDM model, or both. *MusicX* has three main modules:

- (1) *Guidance Control*: The latent optimization is designed to enable conditioning with new modalities, multi-modality, and fine-grained control.
- (2) *Fused Diffusion*: The latent optimization fuses several diffusion trajectories (with guidance control) to enable variable-length conditioned music generation.
- (3) *Fast Distill*: Modified distillation of LDM. The distilled model $f\text{CTM}$ generates music with $T \approx 1 - 10$ steps.

Algorithm 1 Guidance Control

```

1: Input: LDM or  $f\text{CTM}$   $\Phi$ , target  $\mathcal{T}$ , guidance function  $f(\cdot)$ , loss  $l(\cdot, \cdot)$ 
2: Sample seed  $z_T \in \mathcal{N}(0, \mathbb{I})$ 
3: for  $t = T, T-1, \dots, 0$  do
4:   Estimate  $\hat{z}_0$  from data prediction model (Eqn. 3)
5:   Calculate modified noise prediction  $\hat{\epsilon}$  using Eqn. 5
6:   if Further tuning is needed then
7:     Calculate  $\Delta z_0$  using Eqn. 6
8:     Calculate modified noise prediction  $\tilde{\epsilon}$  using Eqn. 7
9:   end if
10:   $z_{t-1} = S(z_t, \tilde{\epsilon}, t)$ 
11: end for

```

2.1. Guidance Control: Multi-modal Conditioning

Figure 1 illustrates the guidance control algorithm in Algorithm 1. Consider a differentiable guidance function f , for example an RNN-based beat estimator, applied to a music sample z , $f(z)$. We also define a corresponding similarity function $l(\cdot, \cdot)$ that measures the closeness of a pair of extracted features. Given a particular target $\mathcal{T}$, the corresponding optimization objective is $l(\mathcal{T}, f(z)) \approx 0$, and we aim to generate a sample z from the music distribution satisfying the constraint.

The choice $f(\cdot)$ is selected from the set of SOTA off-the-shelf models for extracting particular features (Table 1). Classification-guided sampling [30] replaces $\epsilon_\theta(z_t, t)$ in each sampling step $S(z_t, t, \hat{\epsilon})$ with

$$\hat{\epsilon}_\theta(z_t, t) = \epsilon_\theta(z_t, t) - (\sqrt{1 - \alpha_t}) \nabla_{z_t} \log p(c|z_t) \quad (4)$$

This can be replaced with our guidance loss l as,

$$\hat{\epsilon}_\theta(z_t, t) = \epsilon_\theta(z_t, t) - (\sqrt{1 - \alpha_t}) \nabla_{z_t} l(\mathcal{T}, f(\hat{z}_0)) \quad (5)$$

This optimization updates the noise prediction model ϵ_θ . Note that the feature extractor f only operates on clean samples. This would imply that we need to run the diffusion process till $t = 0$ to obtain the clean sample z_0 at each optimization iteration. This is inefficient and would render latent optimization useless. However, the data prediction model of the diffusion sampler predicts the one-step clean sample $\hat{z}_0^1$, which can be used as a proxy for z_0 . Additionally, to ensure stricter adherence to the target $\mathcal{T}$, we can also optimize the data prediction model concurrently with the noise prediction model as,

$$\Delta z_0 = \operatorname{argmin}_{\Delta} l(\mathcal{T}, f(\hat{z}_0 + \Delta)) \quad (6)$$

Then, this Δz_0 is translated back to the noisy data space of z_t by modifying the noise prediction $\hat{\epsilon}$ in Eqn 5.

$$\begin{aligned} \tilde{\epsilon} &= \epsilon_\theta(z_t, t) - \sqrt{\alpha_t / (1 - \alpha_t)} \Delta z_0 \\ z_t &= \sqrt{\alpha_t} (\hat{z}_0 + \Delta z_0) + (\sqrt{1 - \alpha_t}) \tilde{\epsilon} \end{aligned} \quad (7)$$

2.2. Fused Diffusion: Variable Length Guided Generation

We define a fused diffusion sampling process $F_{t-1} = \Psi(F_t, \epsilon_\theta, t) : \bar{A} \times \bar{C} \rightarrow \bar{A}$ as a composition of several vanilla (short) diffusion sampling trajectories $\Phi(z_t, \epsilon_\theta, t) : A \times C \rightarrow A$, where latent input space is $A = R^{H \times W \times C}$ and conditioning space is C for Φ and $\bar{A} = R^{\bar{H} \times \bar{W} \times \bar{C}}$, $\bar{C}$ for Ψ . The mapping between A and $\bar{A}$ spaces is $f_t^i = S_i(F_t)$ and $c_i = S_i(c)$, defined by a segmentation map $S : \bar{A} \rightarrow A$. The segmenter takes a long music clip and splits it into smaller segments. Φ can be

¹We use $\hat{z}_0$ for brevity in place of $\hat{z}_0|z_t$

No.	Conditioning / Control	Feature extractor $f(\cdot)$ /Model	Target $\mathcal{T}$ / Dataset	Loss l
1	Intensity control [18] - time-varying intensity profile	$w \cdot 20 \log(\text{RMS}(V(z_0)))$ with smoothing filter on dB, V is VAE decoder+vocoder	Synthetic target curves	$\ f(z_0) - \mathcal{T}\ _2$ (MSE)
2	Melody control [18] - time-varying pitch progression of music	$\log(C(V(z_0)))$, where C is chromagram (12 pitch classes / STFT)	MAESTRO [19] dataset	$\text{NLL}(f(z_0), \mathcal{T})$ (Negative-Log Likelihood)
3	Structure control [18] - timbre & higher-order music structure	$T(z_0)T(z_0)^T$ (self-similarity matrix from MFCCs without 1st coeff., normalized, smoothed with 2D Savitzky-Golay [20])	Synthetic SS matrix	$\ f(z_0) - \mathcal{T}\ _2$
4	Audio conditioning	Embedding from ImageBind model [21]	MusicCaps dataset (audio targets)	$l = -\cos(f(z_0), \mathcal{T})$
5	Image conditioning	Embedding from ImageBind model [21]	Emotionally paired Music and Image Dataset (EMID) [22] (image targets)	$-\cos(f(z_0), \mathcal{T})$ (cosine similarity)
6	Instrument conditioning - (de)emphasize specific instrument	Probability logits from Wav2Vec2 instrument classifier [23]	Instrument labels (drums, piano, guitar)	$\text{NLL}(f(z_0), \mathcal{T})$
7	Genre conditioning - (de)select specific genre	Probability logits from Wav2Vec2 genre classifier [23]	GTZAN labels [24]	$\text{NLL}(f(z_0), \mathcal{T})$
8	Rhythm control - beat progression	Beat and downbeat activations from RNN beat detector BeatThis [25]	Medley-solos-DB [26] (beat targets)	$\text{NLL}(f(z_0), \mathcal{T})$
9	MIDI conditioning	Embedding from CLAMP3 model [27]	MIDICaps [28] dataset	$-\cos(f(z_0), \mathcal{T})$
10	Sheet music (MXL)	Embedding from CLAMP3 model [27]	Wikifonia [29] dataset	$-\cos(f(z_0), \mathcal{T})$

Table 1: Overview of conditioning and control strategies with corresponding feature extractors, targets, and loss functions.

Model	Time (s)	Objective Metrics			Target Metrics									
		FAD-CLAP ($\downarrow$)	KL-PSST ($\downarrow$)	CLAP-LPAPS ($\downarrow$)	1-MSE ($\downarrow$)	2-NLL ($\downarrow$)	3-MSE ($\downarrow$)	4-COS ($\downarrow$)	5-COS ($\downarrow$)	6-NLL ($\downarrow$)	7-NLL ($\downarrow$)	8-F1 % ($\uparrow$)	9-COS ($\downarrow$)	10-COS ($\downarrow$)
DITTO	82	0.683	1.032	5.826	4.764	0.315	0.0245	0.462	0.157	12.496	11.001	60.03	0.007	0.004
DITTO-2	6	0.644	1.554	7.797	3.311	0.338	0.0220	0.577	0.169	10.635	11.244	58.82	0.001	0.11
AudioLDM2 (ours)	95	0.423	0.894	5.047	3.473	0.192	0.008	0.791	0.669	0.548	0.504	98.16	0.676	0.605
fCTM (ours)	6	0.438	1.076	3.450	3.114	0.252	0.0117	0.769	0.593	0.458	0.499	98.22	0.687	0.531

Table 2: Single Guidance targets: Comparison of objective and target metrics.

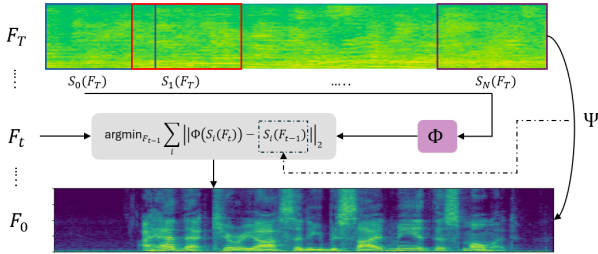


Figure 2: *Fused Diffusion*: A long noise sample F_T is split into standard music segments (10s) $S_i(F_T)$ compatible with LDM/fCTM Φ and predicts $z_{T-1}^{(i)} = \Phi(S_i(F_T))$. The optimization finds a F_{T-1} such that $S_i(F_{T-1})$ matches $z_{T-1}^{(i)}$ subject to guidance control for different conditioning for each segment. The optimization goal is to leverage Φ to generate audio in a potentially different audio space $\bar{A}$ and condition space $\bar{C}$, without any training or finetuning.

$$F_{t-1} = \Psi(F_t | \bar{c}) = \argmin_{F \in \bar{A}} l(F | F_t, \bar{c})$$

$$l(F | F_t, \bar{c}) = \sum_{i=1}^N \|W_i \otimes (S_i(F) - \Phi(S_i(F_t) | c_i))\|^2 \quad (8)$$

where W_i are segment-level weights to ensure cohesion in overlapping regions.

2.3. Fast Distill: Reduced NFEs

LDMs typically need $T \approx 100$ steps (NFEs = 100 for DDIM sampler [15]) to generate good quality music. High-order ODE-based samplers like DPM++ [31] although they reduce NFEs, are limited by truncation errors leading to degradation in quality when $\text{NFE} \leq 5$ and work best at about 15 – 30. On the other hand, Distillation methods use the original LDM model as a teacher and develop a simpler student model that mimics the teacher, but by using reduced NFEs. Distillation methods fall into two categories:

- (1) **Consistency Models (CMs)**: The student learns to generate in one step. The teacher is made to take one step from t to $t - \Delta t$ ($G_\phi(z_t, t, t - \Delta t)$), and then the student’s exponential moving average (EMA) model goes from $t - \Delta t$ to 0, which is compared to the student’s prediction from t to 0 ($G_\theta(z_t, t, 0)$) as seen in Eqn. 9.

- (1) **Consistency Trajectory Models (CTMs)**: The student learns to generate from any timestep to any timestep and the progression is $t \rightarrow s \rightarrow u$ as seen in Eqn. 10.

$$G_\theta(z_t, t, 0) \approx G_{sg(\theta)}(G_\phi(z_t, t, t - \Delta t), t - \Delta t, 0) \quad (9)$$

$$G_\theta(z_t, t, u) \approx G_{sg(\theta)}(G_\phi(z_t, t, s), s, u) \quad (10)$$

The loss is computed based on a distance measure as follows,

$$\mathcal{L}_{CTM}(\theta, \phi) = \mathbb{E}_{t \in [0, T]} \mathbb{E}_{s \in [u, t]} \mathbb{E}_{u \in [0, t]} \mathbb{E}_{z_t | z_0} \mathbb{E}_{\omega \in [0, K]} [d(G_\theta^\omega(z_t, t, u), G_{sg(\theta)}^\omega(G_\phi^\omega(z_t, t, s), s, u))] \quad (11)$$

where ω is the Classifier-free guidance (CFG) guidance scale, i.e., $G_\phi^\omega(z_t, t, s), s, u) = (G_\phi(z_t, t, s), s, u, c) + \omega(G_\phi(z_t, t, s), s, u, c) - G_\phi(z_t, t, s), s, u, \emptyset)$. Both CMs and CTMs student models are trained by picking the guidance scale ω uniformly or at random, and it is usually complicated by its interaction with the distance metric d .

Since CMs are trained using the anytime-to-zero time jump framework, if we want to improve generation quality through multi-step sampling, it requires adding noise to the estimated z_0 to obtain intermediate z_t , but CMs do not learn the diffusion trajectory, as a result, the semantic content of generated samples changes depending on the number of sampling steps, i.e., it leads to accumulation errors. Although CTMs enable flexible multi-step sampling, they require expensive computational resources to fine-tune LDMs, and the time generally exceeds one hundred GPU hours, even reaching the same order of magnitude required for training a LDM from scratch. In addition, incorporating CFG-aware sampling requires additional training and specialized regularization and designs for the distance metric d .

Following [17], we identify the issues and implement three main modifications to CTMs as follows,

- (1) **Mismatch between distillation training and distilled model sampling**: During training, the teacher is sampled repeatedly at timesteps $t \in [0, T]$, but ultimately the student will sample only say 5 fixed timesteps if $\text{NFE} = 5$. This mismatch wastes excessive efforts in the training process that don’t translate to sample quality. The solution is to only train timesteps that will participate in sampling. The key observation is that the gradient field of diffusion is smooth i.e., models exhibit good prediction capabilities at timesteps that they were never trained on as illustrated in [17] Figure 3. Since training at

different timestamps mutually reinforces the model, training on a fine-grained time schedule is unnecessary.

- (2) *Local to Global Trajectory*: In CTMs, the teacher only generates a snippet of the full diffusion trajectory (from t to s) and the loss optimization is independent across different t . Firstly, this makes higher-order ODE solvers impossible to use as they require past sample information. Secondly, the student model doesn't see the full trajectory, and hence, during sampling, similar to CMs, accumulation errors show up at never-before-seen ts . The solution is to allow the teacher to generate the full trajectory from T to 0, and the student mimics the full trajectory. Thus, the student model generates its own trajectories, enabling it to learn to fix the accumulated errors. Since the sample quality of LDMs improves with more steps, a student who properly learns the trajectory performs better at higher NFEs compared to CTMs as illustrated in [17] Figure 8.
- (3) *CFG*: Introducing CFG in distillation training complicates the process. [17] Figure 9 illustrates the effect of a larger guidance scale ω on diffusion trajectories. Thus, it is simpler to perform distillation with $\omega = 1$ and then the student distilled model can be sampled with $\omega \geq 1$.

3. Experiments & Results

We use AudioLDM2 [4] with DPMSolver++(2M) [31] sampler as the base LDM for both optimization and distillation.

3.1. Experiment Setup

3.1.1. Conditioning/Control Targets

Table 1 lists the 10 different modality conditioning/control targets explored in this work. The dataset column lists the dataset from which the targets were selected for each task and similarly, the model column lists the SOTA feature extractor model used.

3.1.2. f CTM Training

DPMSolver++(2M) [31] is the sampler for the teacher AudioLDM2 [4] model and the student f CTM model. We train the student f CTM model for max sampling steps of 10. We use Adam optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.999$, a batch size of 128, and a learning rate of $5e - 5$. The learning rate is divided by 10 halfway through training. Our f CTM is trained with a total of $400K$ teacher trajectories generated (around $3K$ training iterations). $L1$ loss is used as the consistency metric. The training scheme is detailed in [17] (Algorithm 6).

3.1.3. Metrics

We evaluate the generated results using objective metrics: Frechet Audio Distance (FAD) with the CLAP backbone, FAD using OpenL3 embeddings, Kullback-Leibler divergence score using Passt model (KL Passt), Maximum Mean Discrepancy using CLAP embeddings (MMD-CLAP), and CLAP with Learned Perceptual Audio Patch Similarity (CLAP-LPAPS). These metrics are calculated over 2500 generated results across the 10 targets. MusicCAPS [32] is used as the reference for FAD calculation. We also evaluate adherence to the 10 targets via the respective target losses (Table 1). For rhythm, we evaluate with F1 score [33].

Model	2-target (1+2)		2-target (1+3)		3-target (1+2+3)		
	1-MSE ($\downarrow$)	2-NLL ($\downarrow$)	1-MSE ($\downarrow$)	3-MSE ($\downarrow$)	1-MSE ($\downarrow$)	2-NLL ($\downarrow$)	3-MSE ($\downarrow$)
AudioLDM2 (ours)	3.32	0.45	3.27	0.009	3.55	0.49	0.012
f CTM (ours)	4.01	0.56	4.09	0.016	4.21	0.58	0.019

Table 3: *Multiple Guidance targets: Comparison of objective and target metrics.*

3.2. Evaluation

Single Guidance Targets: Results in Table 2 show that our method achieves better performance in terms of objective metrics like FAD, adherence to base TTM model text prompts, and simultaneously achieves better fit to the target guidance/control (e.g., 1-MSE implies MSE for intensity control) as evidenced by great performance across target-based metrics like MSE, Negative Log Likelihood (NLL), Cosine Similarity (COS), etc. This highlights that guidance optimization over multiple timesteps, t and simultaneously optimizing both the noise prediction space ϵ_θ (Eqn. 5) and data prediction space $\hat{z}_0$ (Eqn. 6), offers more robust controllability in terms of both user control and generation quality. Furthermore, introducing the f CTM distilled model achieves comparable generation quality and controllability with 1/5 generation cost (5 steps opposed to 25 of LDM).

Multiple Guidance Targets: Results in Table 3 show that similar performance is achieved when multiple guidance targets are applied simultaneously. We note that our design of f CTM enables further improvements for particularly complex objectives by varying the NFEs or steps to inject more freedom in achieving generation quality and target adherence.

Fused Diffusion: We combine variable-length generation and multiple guidance to generate long music clips with different segments conditioned on different targets.

4. Related Work

■ *Training-based Guidance Methods*: Works like AudioX [34] and Make-An-Audio [35] enable X-to-Audio generation, where X is usually images, audio, or text by re-training or finetuning TTA or TTM diffusion models on large amounts of paired data for each modality. Music works like [36] and [37] are training-based, standalone models that provide symbolic music conditioning and variable-length generation, respectively.

■ *Inference-time Guidance Methods*: Works like DITTO [11] and DITTO-2 [12] offer flexible user controllability without additional training through diffusion initial noise optimization. DITTO-2 [12] replaces the LDM with a Consistency model (CM) that offers one-step generation (NFE=1) at the expense of quality.

5. Limitations & Future Work

We currently support gradient-based optimization and hence, are limited to differentiable guidance losses. To remove this constraint, we hope to explore derivative-free optimization methods like Bayesian Optimization (BO) in the future. The user is responsible for the guidance function and losses for each conditioning or control scheme. We intend to explore using LLMs to translate a user's high-level requirements to guidance parameters. Furthermore, BO would also enable the user to use their perception of the results to directly influence the optimization through Human Feedback-based BO.

6. References

- [1] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, “High-resolution image synthesis with latent diffusion models,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 10 684–10 695.
- [2] J. Ho, W. Chan, C. Saharia, J. Whang, R. Gao, A. Gritsenko, D. P. Kingma, B. Poole, M. Norouzi, D. J. Fleet, and T. Salimans, “Imagen video: High definition video generation with diffusion models,” *arXiv preprint arXiv:2210.02303*, 2022. [Online]. Available: <https://arxiv.org/abs/2210.02303>
- [3] U. Singer, A. Polyak, T. Hayes, X. Yin, J. An, S. Zhang, Q. Hu, H. Yang, O. Ashual, O. Gafni, D. Parikh, S. Gupta, and Y. Taigman, “Make-a-video: Text-to-video generation without text-video data,” *arXiv preprint arXiv:2209.14792*, 2022. [Online]. Available: <https://arxiv.org/abs/2209.14792>
- [4] H. Liu, Y. Yuan, X. Liu, X. Mei, Q. Kong, Q. Tian, Y. Wang, W. Wang, Y. Wang, and M. D. Plumbley, “Audioldm 2: Learning holistic audio generation with self-supervised pretraining,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 32, pp. 2871–2883, 2024.
- [5] K. Chen, Y. Wu, H. Liu, M. Nezhurina, T. Berg-Kirkpatrick, and S. Dubnov, “Musicldm: Enhancing novelty in text-to-music generation using beat-synchronous mixup strategies,” *arXiv preprint arXiv:2308.01546*, 2023. [Online]. Available: <https://arxiv.org/abs/2308.01546>
- [6] K. Wu, Y. Zhang, Y. Liu, Z. Li, Y. Li, Y. Li, and Y. Zhang, “Protein structure generation via folding diffusion,” *Nature Communications*, vol. 15, no. 1, pp. 1–12, 2024. [Online]. Available: <https://www.nature.com/articles/s41467-024-45051-2>
- [7] H. Liu, Z. Chen, Y. Yuan, X. Mei, X. Liu, D. P. Mandic, W. Wang, and M. D. Plumbley, “Audioldm: Text-to-audio generation with latent diffusion models,” *Proceedings of the 40th International Conference on Machine Learning*, vol. 202, pp. 21 450–21 474, 2023. [Online]. Available: <https://proceedings.mlr.press/v202/liu23f.html>
- [8] Z. Evans, J. Parker, C. Carr, Z. Zukowski, J. Taylor, and J. Pons, “Stable audio open,” *arXiv preprint arXiv:2407.14358*, 2024. [Online]. Available: <https://arxiv.org/abs/2407.14358>
- [9] A. Bansal, H.-M. Chu, A. Schwarzschild, S. Sengupta, M. Goldblum, J. Geiping, and T. Goldstein, “Universal guidance for diffusion models,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 1681–1690.
- [10] B. Wallace, A. Gokul, S. Ermon, and N. Naik, “End-to-end diffusion latent optimization improves classifier guidance,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023. [Online]. Available: <https://arxiv.org/abs/2303.13703>
- [11] Z. Novack, J. McAuley, T. Berg-Kirkpatrick, and N. J. Bryan, “Diffusion inference-time t-optimization for music generation,” *arXiv preprint arXiv:2401.12179*, 2024. [Online]. Available: <https://arxiv.org/abs/2401.12179>
- [12] Z. Novack, J. McAuley, T. Berg-Kirkpatrick, and N. Bryan, “Ditto-2: Distilled diffusion inference-time t-optimization for music generation,” *arXiv preprint arXiv:2405.20289*, 2024. [Online]. Available: <https://arxiv.org/abs/2405.20289>
- [13] Y. Song, P. Dhariwal, M. Chen, and I. Sutskever, “Consistency models,” in *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023. [Online]. Available: <https://arxiv.org/abs/2303.01469>
- [14] D. Kim, C.-H. Lai, W.-H. Liao, N. Murata, Y. Takida, T. Uesaka, Y. He, Y. Mitsufuji, and S. Ermon, “Consistency trajectory models: Learning probability flow ode trajectory of diffusion,” *arXiv preprint arXiv:2310.02279*, 2023. [Online]. Available: <https://arxiv.org/abs/2310.02279>
- [15] J. Song, C. Meng, and S. Ermon, “Denoising diffusion implicit models,” *arXiv preprint arXiv:2010.02502*, 2020. [Online]. Available: <https://arxiv.org/abs/2010.02502>
- [16] K. Saito, D. Kim, and T. Shibuya, “Soundctm: Unifying score-based and consistency models for full-band text-to-sound generation,” *arXiv preprint arXiv:2405.18503*, 2024. [Online]. Available: <https://arxiv.org/abs/2405.18503>
- [17] Z. Zhou, D. Chen, C. Wang, C. Chen, and S. Lyu, “Simple and fast distillation of diffusion models,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [18] S.-L. Wu, C. Donahue, S. Watanabe, and N. J. Bryan, “Music controlnet: Multiple time-varying controls for music generation,” *arXiv preprint arXiv:2311.07069*, 2023.
- [19] C. Hawthorne, A. Stasyuk, A. Roberts, I. Simon, C.-Z. A. Huang, S. Dieleman, E. Elsen, J. Engel, and D. Eck, “Enabling factorized piano music modeling and generation with the MAESTRO dataset,” in *International Conference on Learning Representations*, 2019. [Online]. Available: <https://openreview.net/forum?id=11YRjC9F7>
- [20] A. Savitzky and M. J. E. Golay, “Smoothing and differentiation of data by simplified least squares procedures,” *Analytical Chemistry*, vol. 36, no. 8, pp. 1627–1639, 1964.
- [21] R. Girdhar, A. El-Nouby, Z. Liu, M. Singh, K. V. Alwala, A. Joulin, and I. Misra, “Imagebind: One embedding space to bind them all,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2023, pp. 15 180–15 190.
- [22] J. Zou, J. Mei, G. Ye, T. Huai, Q. Shen, and D. Dong, “Emid: An emotional aligned dataset in audio-visual modality,” *arXiv preprint arXiv:2308.07622*, 2023.
- [23] A. Baevski, H. Zhou, A. Mohamed, and M. Auli, “wav2vec 2.0: A framework for self-supervised learning of speech representations,” *arXiv preprint arXiv:2006.11477*, 2020.
- [24] G. Tzanetakis, “Musical genre classification of audio signals,” Ph.D. dissertation, Princeton University, 2002.
- [25] F. Foscari, J. Schlüter, and G. Widmer, “Beat this! accurate beat tracking without DBN postprocessing,” in *Proceedings of the 25th International Society for Music Information Retrieval Conference (ISMIR)*, San Francisco, CA, United States, Nov. 2024.
- [26] V. Lostanlen, C.-E. Cella, R. Bittner, and S. Essid, “Medley-solos-db: a cross-collection dataset for musical instrument recognition,” Zenodo, 2019, version 1.2.
- [27] S. Wu, Z. Guo, R. Yuan, J. Jiang, S. Doh, G. Xia, J. Nam, X. Li, F. Yu, and M. Sun, “Clamp 3: Universal music information retrieval across unaligned modalities and unseen languages,” 2025. [Online]. Available: <https://arxiv.org/abs/2502.10362>
- [28] J. Melechovsky, A. Roy, and D. Herremans, “Midicaps: A large-scale midi dataset with text captions,” *arXiv:2406.02255*, 2024.
- [29] F. Simonetta, “Enhanced wikifonia leadsheet dataset,” 2018, dataset. [Online]. Available: <https://doi.org/10.5281/zenodo.1476555>
- [30] P. Dhariwal and A. Nichol, “Diffusion models beat gans on image synthesis,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. [Online]. Available: <https://arxiv.org/abs/2105.05233>
- [31] C. Lu, Y. Zhou, F. Bao, J. Chen, C. Li, and J. Zhu, “Dpm-solver++: Fast solver for guided sampling of diffusion probabilistic models,” *Machine Intelligence Research*, vol. 22, no. 4, pp. 730–751, 2025.
- [32] A. Agostinelli and T. I. Denk, “Musiclm: Generating music from text,” 2023, introduced MusicCaps dataset (5.5k music-text pairs).
- [33] C. Raffel, B. McFee, E. J. Humphrey, J. Salamon, O. Nieto, D. Liang, D. P. Ellis, and C. C. Raffel, “Mir-eval: A transparent implementation of common mir metrics,” in *ISMIR*, vol. 10, 2014, p. 2014.
- [34] Z. Tian *et al.*, “Diffusion transformer for anything-to-audio generation,” *arXiv preprint arXiv:2503.10522*, 2025. [Online]. Available: <https://arxiv.org/abs/2503.10522>
- [35] R. Huang *et al.*, “Make-an-audio: Text-to-audio generation with prompt-enhanced diffusion models,” in *Proceedings of the 40th International Conference on Machine Learning*, 2023. [Online]. Available: <https://proceedings.mlr.press/v202/huang23i.html>

- [36] G. Mittal *et al.*, “Symbolic music generation with diffusion models,” in *Proceedings of the 22nd International Society for Music Information Retrieval Conference*, 2021. [Online]. Available: <https://archives.ismir.net/ismir2021/paper/000058.pdf>
- [37] Z. Evans *et al.*, “Long-form music generation with latent diffusion,” *arXiv preprint arXiv:2404.10301*, 2024. [Online]. Available: <https://arxiv.org/abs/2404.10301>